

SESSION 4

RAD-541 Magnetic Resonance Imaging II

K-Space Filling

How encoded MR signals are organized before image reconstruction

Course	Magnetic Resonance Imaging II
Date	Sept, 2026
Lecturer	A. A. Zubair
Suggested study time	240–300 minutes

Learning Objectives

By the end of this session, you should be able to:

1. define k-space in simple terms and distinguish it from image space;
2. explain the meaning of the k_x and k_y axes in conventional 2D Cartesian MRI;
3. describe how one readout samples one line of k-space;
4. explain how repeated phase-encoding amplitudes fill successive k-space lines;
5. distinguish low spatial frequencies from high spatial frequencies;
6. explain why the centre of k-space strongly influences overall signal and contrast while the periphery contributes strongly to edges and fine detail;
7. relate k-space sampling to field of view, matrix, spatial resolution and scan time;
8. describe common Cartesian filling orders such as linear and centric ordering at a conceptual level; and
9. identify common misconceptions about what an individual k-space point represents.

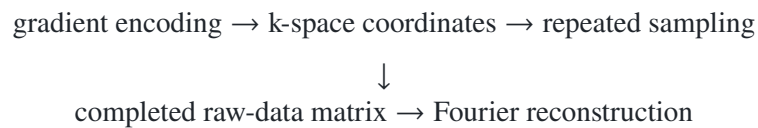
The central idea

K-space is the mathematical array in which MRI stores **raw complex spatial-frequency samples** after gradient encoding. It is **not a picture of the patient**. A single k-space sample does not belong to one anatomical point; it contributes information to the entire reconstructed image. After the required samples have been collected, Fourier transformation converts their spatial-frequency and phase information into image space.

1 Session Orientation

Sessions 2 and 3 explained how gradients create spatial labels: slice selection chooses the section, frequency encoding creates a position-dependent frequency pattern during readout, and phase encoding leaves a position-dependent phase pattern before readout. Session 4 asks the next question: *where are all of those encoded measurements placed before they become an image?*

A useful map is



The most important shift in thinking is that k-space is organized by **spatial frequency**, not by anatomy. Its centre and periphery therefore describe different scales of image variation rather than different parts of the body.

Why this matters at the console

Choices such as FOV, phase matrix, readout matrix, partial sampling, acceleration and acquisition order all change how k-space is sampled. Understanding the underlying map helps the radiographer predict effects on resolution, scan time, aliasing and contrast rather than memorizing isolated parameter rules.

2 Why MRI needs k-space

In Session 3, frequency encoding assigned one in-plane coordinate and phase encoding assigned the other. The receiver coil then detected a complex mixture of signals from the whole selected slice. MRI still needs an organized place to store those encoded measurements before reconstruction. That place is **k-space**.

MRI does not usually acquire one pixel at a time. Instead, each receiver sample is a complex measurement containing combined contributions from many locations in the selected slice. Spatial encoding changes the phase and frequency relationships within that mixture, and k-space provides an organized coordinate system for storing the resulting measurements before reconstruction.

K-space is not anatomy

The top of k-space does not correspond to the top of the patient, and the centre of k-space does not correspond to the centre of the body. K-space coordinates describe **spatial frequency content**, not anatomical position.

3 What is a spatial frequency?

Spatial frequency describes how rapidly image signal changes with distance. This is different from Larmor frequency, which describes how rapidly magnetization rotates with time.

Consider two simple image patterns:

- a large smooth region that changes slowly from bright to dark has predominantly **low spatial-frequency** content;
- many fine alternating bright and dark structures contain more **high spatial-frequency** content.

Low spatial frequencies therefore describe broad shapes, overall brightness and slowly varying contrast. High spatial frequencies describe rapid signal changes such as edges, fine boundaries and small structural detail.

Spatial-frequency range	Image behaviour it represents	Typical visual contribution
Low	Signal changes slowly over distance	Broad shape, overall intensity and slowly varying contrast
High	Signal changes rapidly over short distances	Edges, fine boundaries and small structural detail

Do not confuse two meanings of “frequency”

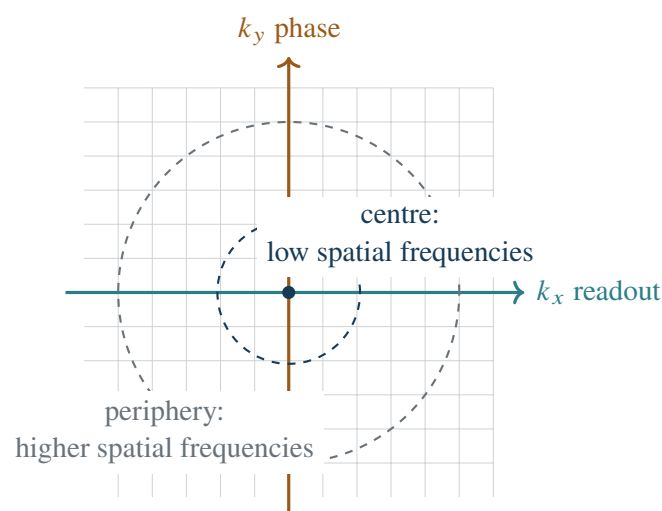
Larmor frequency is a temporal frequency measured in hertz and depends on magnetic field strength. **Spatial frequency** describes how rapidly image information changes across distance. K-space is organized by spatial frequency, not by anatomical distance.

4 The axes of 2D k-space

For a conventional 2D Cartesian acquisition, k-space is commonly drawn as a rectangular grid with two principal axes:

- k_x – usually associated with the frequency/readout direction;
- k_y – usually associated with the phase-encoding direction.

These axes do not represent centimetres or millimetres in the patient. They represent spatial-frequency coordinates. Values close to the origin ($k_x = 0, k_y = 0$) represent low spatial frequencies. Values farther from the origin represent progressively higher spatial frequencies.



5 How gradients move the acquisition through k-space

The bridge between gradient physics and k-space is the **gradient area**. In one dimension, a useful relationship is

$$k(t) = \bar{\gamma} \int_0^t G(\tau) d\tau,$$

where $\bar{\gamma} = \gamma/2\pi$ is in Hz/T and $G(t)$ is the gradient waveform. With this convention, k is expressed in cycles per metre.

This equation is more useful conceptually than mathematically at this stage:

- a gradient applied for longer, or with greater amplitude, produces a larger change in k-space position;
- reversing gradient polarity moves the sampling position in the opposite k-space direction; and
- when the net gradient area returns to zero along an axis, the sampling coordinate returns to $k = 0$ on that axis.

For conventional Cartesian imaging, the phase-encoding gradient establishes the required k_y position before readout, while the readout gradient then drives the trajectory across k_x as the echo is sampled.

Gradient area, not gradient amplitude alone

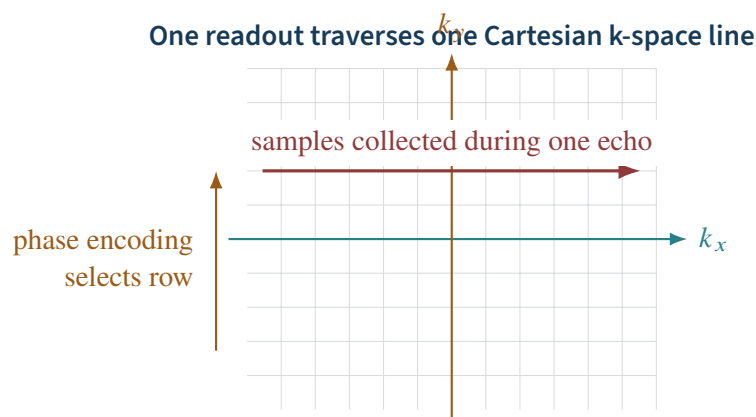
K-space location depends on the **time integral of the gradient**. A short strong gradient and a longer weaker gradient can therefore produce the same k-space displacement if their areas are equal.

6 How one echo fills one line of k-space

In conventional Cartesian 2D imaging, one repetition commonly acquires one horizontal line of k-space. The phase-encoding gradient establishes the vertical k_y position of that line. A preparatory readout lobe may first move the trajectory toward one side of k_x ; the readout gradient then carries the sampling point across the line while the receiver digitizes the echo. The point $k_x = 0$ is crossed near the middle of the readout and, in many conventional sequences, is arranged close to the echo maximum.

The sequence can be understood as four steps:

1. Apply one chosen phase-encoding gradient amplitude.
2. Begin readout with the frequency-encoding gradient.
3. Sample the echo repeatedly from one side of k_x through the centre to the other side.
4. Store those samples along one k-space row.



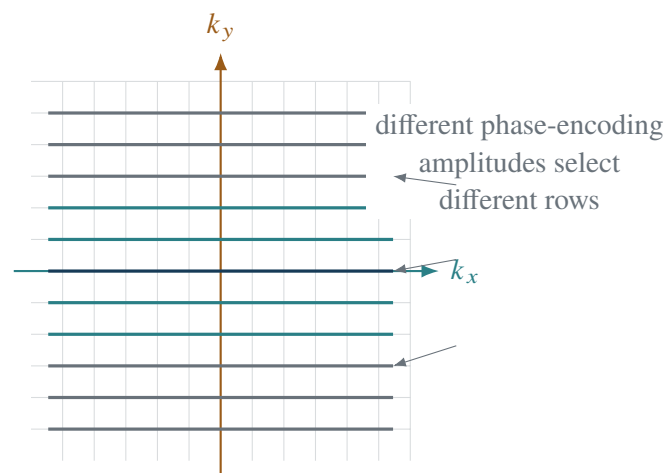
Why this matters to the radiographer

If the frequency matrix contains 256 readout samples, one echo may provide approximately 256 samples across a single k-space line. But the scanner still needs many different phase-encoding steps to fill the other rows. This is one reason the phase matrix is strongly linked to acquisition time in conventional Cartesian imaging.

7 How repeated phase encoding fills the grid

The phase-encoding gradient is changed from one repetition to the next. A small phase-encoding moment places data near the centre of the k_y axis, whereas larger positive or negative phase-encoding moments place data progressively farther into the upper or lower parts of k-space.

Suppose a sequence uses 256 phase-encoding steps. Conceptually, the pulse sequence is repeated many times, with a different phase-encoding amplitude each time. Each repetition therefore fills another row until the required k-space matrix has been sampled.



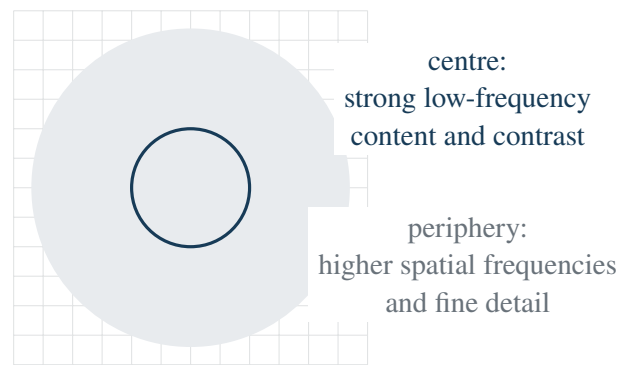
One readout line, many repetitions

In conventional Cartesian 2D MRI, the readout gradient samples across k_x during each echo, while repeated phase-encoding steps move the acquisition to different k_y rows. A useful simplification is: **readout fills across a line; phase encoding chooses which line.**

8 Why the centre of k-space is so important

The central region of k-space contains low spatial-frequency information. These components describe broad signal distributions: overall intensity, large structures and slowly varying contrast. The exact origin, $k_x = k_y = 0$, is sometimes called the **DC component**; conceptually, it represents the spatially non-varying component of the signal.

The centre often carries high signal amplitude and has a powerful influence on image appearance. In many pulse sequences, the timing of the echo centre is coordinated so that central k-space is sampled near the desired echo time. If the central lines are acquired under different contrast conditions – for example during a changing contrast bolus or after a preparation pulse – the reconstructed image can show markedly different contrast even when peripheral sampling is similar.



The centre does not contain “all the contrast” and the edge does not contain “all the resolution”

These statements are useful teaching shortcuts, but the final image depends on the entire k-space dataset. Central k-space strongly influences broad contrast and signal; peripheral k-space strongly contributes to edges and fine spatial detail. They work together rather than acting as completely separate images.

9 What happens if we keep only part of k-space?

A powerful way to understand k-space is to imagine reconstructing an image from selected regions only.

9.1 Keeping mainly the centre

If mostly central k-space is retained, low spatial frequencies remain while many high spatial frequencies are removed. The resulting image preserves broad anatomy and contrast but becomes blurred because sharp boundaries and fine detail have been lost.

9.2 Keeping mainly the periphery

If mostly peripheral k-space is retained, the broad low-frequency signal is greatly reduced. The reconstructed image emphasizes edges and rapid changes rather than normal overall anatomy and contrast.

A useful reconstruction thought experiment

Keeping mainly central k-space produces a recognizable but blurred image because broad low-frequency information remains while fine detail is lost. Keeping mainly peripheral k-space suppresses much of the broad signal and emphasizes edges. Neither region alone represents a complete diagnostic image.

10 K-space and spatial resolution

Spatial resolution is related to how far k-space is sampled into high spatial frequencies. Sampling farther from the centre means the acquisition captures more rapidly varying spatial information, allowing smaller structures and sharper edges to be represented.

In simple Cartesian terms, increasing the acquired matrix at the same field of view usually means collecting more independent spatial-frequency samples. This can reduce pixel size and improve potential spatial resolution, although SNR, acquisition time and other sequence constraints must also be considered.

A useful conceptual relationship is:

$$\Delta x \approx \frac{FOV_x}{N_x}, \quad \Delta y \approx \frac{FOV_y}{N_y}.$$

The k-space view of the same idea separates two concepts:

$$\Delta k \approx \frac{1}{FOV}, \quad k_{\max} \approx \frac{1}{2\Delta x}.$$

Thus, the **spacing** between samples mainly determines the supported field of view, whereas the **maximum distance sampled from the centre** determines the finest spatial detail that can be represented. At fixed FOV, increasing matrix size keeps Δk approximately unchanged but extends sampling farther toward larger $|k|$, improving nominal pixel resolution at the cost of smaller voxels and, for added phase lines, longer acquisition time.

Matrix is not a free improvement

Increasing phase matrix at unchanged FOV usually increases the number of phase-encoding steps. In a conventional sequence this can lengthen scan time and reduce voxel volume/SNR. The radiographer therefore balances resolution against SNR, motion risk and examination time.

11 K-space spacing and field of view

The distance between neighbouring k-space samples is often written as Δk . Conceptually, tighter spacing in k-space corresponds to a larger field of view in image space, whereas wider spacing corresponds to a smaller field of view.

This inverse relationship helps explain aliasing. If the field of view is too small for the anatomy being encoded, the object extends beyond the supported image-space region and may wrap into the displayed image.

FOV is an inverse k-space relationship

Smaller Δk means more closely spaced k-space samples and supports a larger FOV. Larger Δk supports a smaller FOV. If anatomy extends beyond that supported FOV, its signal can be mapped back into the image as aliasing or wrap-around.

12 Linear versus centric filling order

K-space lines do not always have to be acquired in the same order. Two useful conceptual examples are:

12.1 Linear ordering

The scanner progresses through the k-space in an orderly sequence from one side to the other. Central lines are therefore acquired somewhere near the middle of the acquisition.

12.2 Centric ordering

The central k-space lines are acquired early, followed by progressively more peripheral data. This can be useful when the desired image contrast is strongest or most time-sensitive immediately after a preparation pulse or contrast event.

Ordering	How lines are acquired	Why timing can matter
Linear	Progresses through k_y in an ordered sequence	Central data are acquired near the middle of the ordering
Centric	Acquires central lines early, then moves outward	Useful when desired contrast is strongest immediately after preparation or during a time-sensitive enhancement phase

Order affects when contrast is encoded

The order of k-space filling can affect image contrast when signal changes during the acquisition. Central k-space is especially important because it carries strong low-spatial-frequency signal and contrast information. This becomes important in contrast-enhanced MRI and many fast-sequence strategies.

13 Other K-Space Trajectories

Cartesian line-by-line sampling is the clearest way to learn basic k-space, but MRI can also traverse k-space along radial spokes, spirals and other trajectories. These approaches use different gradient waveforms and can alter speed, motion behaviour and artifact appearance. The important undergraduate principle is that **k-space is a coordinate system that can be sampled along different paths**; the Fourier reconstruction must account for how those samples were acquired.

14 A complete 2D Cartesian filling sequence

The entire process can now be followed from one repetition to the next:

1. Select the required slice using RF excitation and the slice-select gradient.
2. Apply one phase-encoding gradient amplitude.
3. Rephase as required by the pulse sequence.
4. Apply the frequency/readout gradient and sample the echo.
5. Store the samples across one k_x line at the selected k_y value.
6. Repeat the sequence with a new phase-encoding amplitude.
7. Continue until the required set of k-space lines has been filled.
8. Send the completed data for Fourier reconstruction into image space.

The complete loop

For basic 2D Cartesian imaging: **select the slice → choose one k_y position with phase encoding → read across k_x while sampling the echo → store the line → repeat with a new phase step**. When the required matrix is complete, reconstruction converts k-space into image space.

Why a larger phase matrix takes longer

Suppose a basic spin-echo acquisition has $TR = 500$ ms and requires 128 phase-encoding lines. Ignoring acceleration, averages and other sequence details, a rough acquisition-time estimate is

$$TA \approx TR \times N_{phase} = 0.5 \times 128 = 64 \text{ s.}$$

If the phase matrix is increased to 256 with everything else unchanged,

$$TA \approx 0.5 \times 256 = 128 \text{ s.}$$

The scan time approximately doubles because twice as many phase-encoding repetitions are needed.

Key Takeaways

- K-space stores spatially encoded raw MR data; it is not an anatomical image.
- In conventional Cartesian 2D MRI, the phase-encoding gradient chooses the k_y row and the readout gradient samples across k_x .
- Repeated phase-encoding amplitudes fill successive k-space lines.
- Central k-space contains low spatial frequencies that strongly influence broad signal and contrast; peripheral k-space contains higher spatial frequencies that strongly contribute to edges and fine detail.
- Gradient area determines movement through k-space: phase encoding selects k_y , while readout traverses k_x .
- $\Delta k \approx 1/FOV$, whereas larger k_{max} supports finer spatial resolution.
- K-space can be filled in different orders and trajectories; ordering matters when signal or contrast changes during acquisition.

Key terms: k-space; image space; spatial frequency; k_x ; k_y ; gradient area; Cartesian sampling; readout line; central k-space; DC component; peripheral k-space; low spatial frequency; high spatial frequency; Δk ; k_{max} ; linear ordering; centric ordering.

Self-Assessment Questions

1. Why is k-space not an anatomical map of the patient?
2. What do k_x and k_y represent in conventional 2D Cartesian imaging?
3. Explain how gradient area determines k-space position and how one readout fills one Cartesian line.
4. Why must the phase-encoding gradient be changed from repetition to repetition?
5. What image information is strongly associated with central versus peripheral k-space?
6. How does phase matrix affect acquisition time in a simple Cartesian sequence?
7. State the approximate relationships between Δk and FOV, and between k_{max} and spatial resolution.
8. Why might a sequence use centric rather than linear ordering?

Series navigation

Next session: Fourier Transformation & Complete MRI Image Formation. Session 5 will show how the spatial-frequency information organized in k-space is transformed into image space, completing the path from B_0 and gradient encoding to the reconstructed MR image.

